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# Multi-Agent Reinforcement Learning for Local Energy Markets: DSO Price Optimization with Flexibility

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## Introduction

With the growing penetration of distributed renewable resources, local energy markets (LEMs) emerge as a key paradigm for coordinating heterogeneous actors (producers, consumers, and prosumers) within constrained distribution grids. Ensuring both efficiency and reliability is challenging, as individual optimization often conflicts with global objectives, particularly under congestion.

While distributed optimization methods such as ADMM provide rigorous coordination mechanisms, they lack adaptability to uncertainty and evolving agent behavior. Multi-Agent Reinforcement Learning (MARL) offers a complementary approach, enabling agents to learn adaptive strategies that align local incentives with system-wide stability.

A central challenge lies in the role of the Distribution System Operator (DSO), whose primary lever to influence the market is dynamic pricing. Designing price signals that simultaneously mitigate congestion and activate flexibility resources represents a critical step toward resilient and fair LEM operation.

This project therefore focuses on MARL-based DSO price optimization with flexibility, extending prior work on learning-based congestion management to incorporate storage and demand response. The objective is to develop adaptive and scalable pricing strategies that ensure efficient, grid-compliant coordination in decentralized energy systems.

## 2 Objectives

The primary objective of this work was to investigate how **multi-agent reinforcement learning (MARL)** can be integrated into **local energy markets (LEMs)** to enhance congestion management in distribution networks. Specifically, the research sought to:

- Develop a **distributed optimization framework** based on the **Alternating Direction Method of Multipliers (ADMM)** for coordinating autonomous agents under network constraints.
- Design and implement a **learning-based control mechanism** using reinforcement learning (RL) to dynamically adjust price signals in order to mitigate grid congestion.
- Explore **reward function design** as a means to improve the robustness of the RL agent, particularly under heterogeneous and dynamic agent behavior.
- Lay the foundation for **future integration of flexibility resources** (e.g., battery storage) as additional RL agents, enabling joint optimization of energy costs and grid stability.

## 3 Progress

The study progressed through three major stages:

### 1. Centralized to Distributed Market Modeling

The initial step involved formulating the LEM as an optimization problem under **convex cost functions and symmetric exchange constraints**. This problem was decomposed using **ADMM**, enabling **peer-to-peer coordination** through local computations and global synchronization. Comparative analysis confirmed the equivalence of centralized and distributed

solutions, while demonstrating improved convergence speed in the multi-agent setup due to explicit consensus enforcement.

### 2. Integration of Reinforcement Learning for Congestion Management

A **Soft Actor-Critic (SAC)** agent was trained to generate **dynamic price signals** influencing agent decisions. Multiple **reward function designs** were tested:

- An initial formulation penalizing constraint violations beyond the threshold.
- A **Gaussian-shaped continuous reward** centered on an optimal operating point, which yielded significant improvements in stability and congestion reduction ( $\approx 23.4\%$  improvement compared to the initial reward).

### 3. Deployment in a Multi-Agent Environment

Despite differences in observation spaces between centralized and distributed configurations, the controller maintained effectiveness, reducing congestion violations by approximately **90% compared to a baseline without RL control**.

## 4 Scope and Goals

### 4.1 General Scope and Main Objectives

This work primarily addressed **congestion management** in **peer-to-peer energy markets** through the combination of **distributed optimization** and **learning-based control mechanisms**.

The scope was deliberately restricted to:

- A small-scale LEM comprising a producer, consumer, and DSO, with a **community manager agent** for global synchronization (Figure 1).

- Price-based coordination as the sole lever for influencing agent behavior, with price signals applied to all exchanges between agents, not just with the DSO, without any direct control over peer-to-peer transactions.
- Evaluation of **reward shaping strategies** to enhance RL policy robustness in static and dynamic environments.

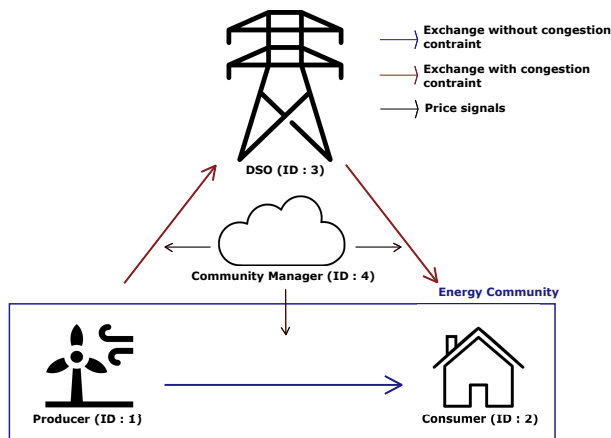


Figure 1: *Simplified Local Energy Market under Study*

The next research phase aims to introduce **flexibility through a controllable battery system**, operated by a dedicated RL agent. The key objective is to design a policy enabling the battery to **charge when DSO exchange prices are low and discharge when they are high**, based on dynamic price signals. This mechanism would:

- Reduce energy costs for market participants through optimal storage operation.
- Smooth demand profiles and mitigate peaks, improving overall grid stability.
- Enhance the adaptability and scalability of price-based coordination in realistic environments.

Ultimately, the integration of such flexibility resources will pave the way for a **multi-agent learning architecture** where:

- A DSO-controlled agent manages congestion through price signals.
- A battery agent optimizes storage for cost and stability objectives.
- Both agents interact within a decentralized market governed by fairness and efficiency principles.

## 4.2 Specific Achievements and Future Research Directions

### Implementation Status and Current Achievements

- The battery integration within the peer-to-peer market has been successfully implemented.
- Multi-step simulations have been performed, validating the basic dynamics of storage operation and its interactions with other agents.
- The current setup provides a flexible framework that can be modified to test different configurations and scenarios.

### Future Objectives

- Calibrate the environment to ensure proper dimensioning of interactions, enabling meaningful RL training.
- Train the RL agent to optimally control price signals while managing the battery, aiming at:
  - Cost-efficient operation for market participants.
  - Smoother demand profiles and congestion mitigation.
  - Enhanced stability and scalability of the decentralized market.
- Refine the learning-based control mechanism to coordinate storage and pricing decisions in a multi-agent setting, ensuring fairness, efficiency, and grid reliability.